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Information notification method and information notification device

Abstract

An information notification method includes: obtaining a plurality of incident information items on a plurality of incidents caused by cyber attacks on a monitoring target during a first period in past; determining, based on the plurality of incident information items obtained, for each type of the plurality of incidents, an incident trend of incidents of the type in the monitoring target during a second period in the future and a priority of a provisional response or a permanent response to an incident of the type; and notifying of the incident trend and the priority that are determined for each type of the plurality of incidents.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This is a continuation application of PCT International Patent Application No. PCT/JP2023/044743 filed on Dec. 13, 2023, designating the United States of America, which is based on and claims priority of Japanese Patent Application No. 2022-202232 filed on Dec. 19, 2022 and Japanese Patent Application No. 2023-206062 filed on Dec. 6, 2023.

FIELD

(1) The present disclosure relates to an information notification method and an information notification device.

BACKGROUND

(2) Patent Literature (PTL) 1 discloses an alert notification device that performs cluster analysis based on characteristics of incidents that occur, and determines a high priority for an incident that is more seemed to be an anomaly, based on a result of the analysis, and notifies of an alert.

CITATION LIST

Patent Literature

(3) PTL 1: Japanese Unexamined Patent Application Publication No. 2019-175070

SUMMARY

(4) However, the technique according to PTL 1 can be improved upon. In view of this, the present disclosure provides an information notification method or the like capable of improving upon the above related art.

(5) An information notification method according to an aspect of the present disclosure includes: obtaining a plurality of incident information items on a plurality of incidents caused by cyber attacks on a monitoring target during a first period in past; determining, based on the plurality of incident information items obtained, for each type of the plurality of incidents, an incident trend of incidents of the type in the monitoring target during a second period in future and a priority of a provisional response or a permanent response to an incident of the type; and presenting the incident trend and the priority that are determined for each type of the plurality of incidents.

(6) It should be noted that general or specific aspects of the present disclosure may be realized as a system, a device, an integrated circuit, a computer program, a non-transitory computer-readable recording medium such as a CD-ROM, or any given combination thereof.

(7) According to the above-described aspect, an information notification method or the like according to the present disclosure is capable of improving upon the above related art.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) These and other advantages and features of the present disclosure will become apparent from

the following description thereof taken in conjunction with the accompanying drawings that illustrate a specific embodiment of the present disclosure.

- (2) FIG. 1 is a schematic view of an information notification system according to an embodiment.
- (3) FIG. 2 is a block diagram illustrating an example of a hardware configuration of an information notification device according to the embodiment.
- (4) FIG. 3 is a block diagram illustrating an example of a hardware configuration of a vehicle according to the embodiment.
- (5) FIG. 4 is a block diagram illustrating an example of a functional configuration of the information notification system according to the embodiment.
- (6) FIG. 5 is a table showing a specific example of incident statistical information.
- (7) FIG. 6 is a table showing another example of the incident statistical information.
- (8) FIG. 7 is a table showing another example of the incident statistical information.
- (9) FIG. 8 is a table showing another example of the incident statistical information.
- (10) FIG. 9 is a graph illustrating change over time in occurrence count of incident information items.
- (11) FIG. 10 is a sequence diagram illustrating an example of an information notification method executed by the information notification system according to the embodiment.
- (12) FIG. 11 is a sequence diagram illustrating an example of an information notification method executed by an information notification system according to Variation 1.
- (13) FIG. 12 is a sequence diagram illustrating an example of an information notification method executed by an information notification system according to Variation 2.
- (14) FIG. 13 is a sequence diagram illustrating an example of an information notification method executed by an information notification system according to Variation 3.
- (15) FIG. 14 is a graph illustrating an example of a probability density function with a random variable that is an index indicating an incident trend.
- (16) FIG. 15 is a diagram illustrating an example in which a distribution is given by a range including a plurality of index values that can take on in the index indicating the incident trend.
- (17) FIG. 16 is a diagram illustrating an example in which the distribution is given by the plurality of index values that can take on in the index indicating the incident trend.
- (18) FIG. 17 is a diagram for describing a first example of a priority determination method in Variation 4.
- (19) FIG. 18 is a diagram for describing a second example of the priority determination method in Variation 4.
- (20) FIG. 19 is a diagram for describing a third example of the priority determination method in Variation 4.
- (21) FIG. 20 is a diagram for describing a fourth example of the priority determination method in Variation 4.
- (22) FIG. 21 is a diagram for describing a fifth example of the priority determination method in Variation 4.
- (23) FIG. 22 is a diagram for describing a sixth example of the priority determination method in Variation 4.
- (24) FIG. 23 is a diagram for describing a seventh example of the priority determination method in Variation 4.
- (25) FIG. 24 is a diagram for describing an eighth example of the priority determination method in Variation 4.
- (26) FIG. 25 is a sequence diagram illustrating an example of an information notification method executed by an information notification system according to Variation 4.

DESCRIPTION OF EMBODIMENT

(27) (Underlying Knowledge Forming Basis of the Present Disclosure)

(28) The main purpose of the technique according to PTL 1 is to give a high priority to an incident

that is more seemed to be an anomaly, out of incidents that are notified of in large quantity, and thus the technique cannot give a high priority to incidents that cause damage expected to escalate in the future. This fact is described in detail below.

(29) When an information security vulnerability is found in a piece of equipment such as a vehicle, a plurality of cyber attacks of the same type may occur in the piece of equipment or another piece of equipment of the same model as the piece of equipment. Therefore, to minimize damage caused by the cyber attacks, a security incident response team (SIRT) should determine a provisional response or a permanent response in an early stage and respond to the cyber attacks.

(30) However, the provisional response is, for example, fail soft (breakdown), which has a significant impact on end users. Thus, it is difficult to determine to perform a provisional response as a response to a cyber attack in a situation where information for decision-making is insufficient. In addition, in a permanent response, it is difficult to respond to a plurality of incidents simultaneously with a limited number of members of a SIRT.

(31) As a result of intensive studies, the inventors have found an information notification method and the like capable of providing information for decision-making to make it accurate and easy to determine whether to carry out the provisional response and determination of a response priority of the permanent response.

(32) An information notification method according to a first aspect of the present disclosure includes: obtaining a plurality of incident information items on a plurality of incidents caused by cyber attacks on a monitoring target during a first period in past; determining, based on the plurality of incident information items obtained, for each type of the plurality of incidents, an incident trend of incidents of the type in the monitoring target during a second period in future and a priority of a provisional response or a permanent response to an incident of the type; and presenting the incident trend and the priority that are determined for each type of the plurality of incidents.

(33) Therefore, it is possible to provide information for decision-making to make it accurate and easy to determine whether to carry out a provisional response and determination of a response priority of a permanent response. For example, when a SIRT receives a notification, the SIRT can execute a provisional response or a permanent response in an early stage and can easily determine whether to carry out the provisional response and a priority of carrying out the permanent response. Accordingly, it is possible to preferentially respond to an incident that will have a great impact on security even with a limited number of members.

(34) An information notification method according to a second aspect of the present disclosure is the information notification method according to the first aspect, wherein the priority being a higher priority indicates that the provisional response or the permanent response is made earlier, and based on the incident trend, the priority is determined to be higher with an increase in potential future damage caused by a cyber attack.

(35) Accordingly, it is possible to preferentially respond to an incident in which a cyber attack will cause great potential future damage even with a limited number of members.

(36) An information notification method according to a third aspect of the present disclosure is the information notification method according to the second aspect, wherein the incident trend relates to change over time in occurrence count of incidents for each type of the plurality of incidents.

(37) Therefore, for example, when a SIRT receives the notification, the SIRT can easily grasp a future incident trend of incident information.

(38) An information notification method according to a fourth aspect of the present disclosure is the information notification method according to the third aspect, wherein the incident trend is a slope of the change over time at a first time point in the second period, and the priority is determined to be higher with an increase in the slope.

(39) Accordingly, a priority of a response is determined to be higher for an incident that has a steeper slope and has an increasing tendency. Thus, it is possible to promote a more prioritized

response for an incident that has a steeper slope and has an increasing tendency.

(40) An information notification method according to a fifth aspect of the present disclosure is the information notification method according to the third aspect, wherein the incident trend is a curve illustrating the change over time during the second period, and the priority is determined to be higher with an increase in a maximum value of the curve.

(41) Accordingly, a priority of a response is determined to be higher for an incident that has a curve with a larger maximum value. Thus, it is possible to promote a more prioritized response for an incident that has a curve with a larger maximum value.

(42) An information notification method according to a sixth aspect of the present disclosure is the information notification method according to the third aspect, wherein the incident trend is a curve illustrating the change over time during the second period, and the priority is determined to be higher with an increase in a time during which the curve is greater than a given occurrence count.

(43) Accordingly, a priority of a response for an incident having a longer time during which its curve is greater than the given occurrence count is determined to be higher. Thus, it is possible to promote a more prioritized response for an incident having a longer time during which its curve is greater than the given occurrence count.

(44) An information notification method according to a seventh aspect of the present disclosure is the information notification method according to the third aspect, wherein the incident trend is a total count of occurrence counts of incidents during the second period, and the priority is determined to be higher with an increase in the total count.

(45) Accordingly, a priority of a response is determined to be higher for an incident that causes a higher total count of occurrence counts of incident information items during the second period. Thus, it is possible to promote a more prioritized response for an incident that causes a higher total count of occurrence counts of incident information items during the second period.

(46) An information notification method according to an eighth aspect of the present disclosure is the information notification method according to the third aspect, wherein the incident trend is a total count of occurrence counts of incidents during the second period, and the priority is determined to be higher with an increase in a total amount of damage that is calculated by multiplying the total count by an amount of damage that occurs when one incident occurs.

(47) Accordingly, a priority of a response is determined to be higher for an incident that will result in a larger total amount of damage. Thus, it is possible to promote a more prioritized response for an incident that will result in a larger total amount of damage.

(48) An information notification method according to a ninth aspect of the present disclosure is the information notification method according to any one of the first to eighth aspects, wherein the incident trend includes a distribution of a plurality of index values that are likely to be determined in an index indicating the incident trend in the determining.

(49) Therefore, it is possible to determine a priority with consideration given to a likelihood of prediction of an incident trend.

(50) An information notification method according to a tenth aspect of the present disclosure is the information notification method according to the ninth aspect, wherein the distribution is given by a probability density function with a random variable that is the index indicating the incident trend.

(51) Therefore, it is possible to determine a priority with consideration given to an occurrence probability of an incident trend.

(52) An information notification method according to an eleventh aspect of the present disclosure is the information notification method according to the ninth aspect, wherein the distribution is given by a first range that includes the plurality of index values. Therefore, it is possible to determine a priority with consideration given to a distribution of an incident trend.

(53) An information notification method according to a twelfth aspect of the present disclosure is the information notification method according to any one of the ninth to eleventh aspects, wherein the priority is determined to be higher with an increase in an expectation based on the distribution.

(54) An information notification method according to a thirteenth aspect of the present disclosure is the information notification method according to any one of the ninth to eleventh aspects, wherein the priority is determined to be higher with an increase in a maximum value of the plurality of index values.

(55) An information notification method according to a fourteenth aspect of the present disclosure is the information notification method according to any one of the ninth to eleventh aspects, wherein the priority is determined to be higher with an increase in a variance in the distribution.

(56) An information notification method according to a fifteenth aspect of the present disclosure is the information notification method according to any one of the ninth to eleventh aspects, wherein the priority is determined to be higher with an increase in a proportion that is calculated based on the distribution and is accounted for by a distribution of index values greater than or equal to a first threshold value in the index.

(57) An information notification method according to a sixteenth aspect of the present disclosure is the information notification method according to any one of the ninth to eleventh aspects, wherein the priority is determined using a machine learning model generated by machine learning using, as an input, a plurality of incident trends corresponding to a plurality of first incident information items that are issued during a third period in the past and using, as ground truth data, a first evaluation value determined for each of the plurality of incident trends.

(58) An information notification method according to a seventeenth aspect of the present disclosure is the information notification method according to any one of the ninth to eleventh aspects, wherein a precedence based on the priority is determined using a machine learning model generated by machine learning using, as an input, a plurality of incident trends corresponding to a plurality of first incident information items that are issued during a third period in the past and using, as ground truth data, a precedence determined for each of the plurality of incident trends.

(59) An information notification method according to an eighteenth aspect of the present disclosure is the information notification method according to any one of the ninth to eleventh aspects, wherein the determining further includes: calculating at least two second evaluation values for each of the plurality of incidents by ranking the plurality of incidents according to each of at least two values among: an expectation based on the distribution; a maximum value of the plurality of index values; a variance in the distribution; a proportion that is calculated based on the distribution and is accounted for by a distribution of index values greater than or equal to a first threshold value in the index; and an evaluation value determined using a machine learning model based on a plurality of incident trends corresponding to a plurality of first incident information items that are issued during a third period in the past; and calculating a third evaluation value for each of the plurality of incidents by adding up the at least two second evaluation values, and the priority is determined to be higher with an increase in the third evaluation value.

(60) An information notification method according to a nineteenth aspect of the present disclosure is the information notification method according to any one of the ninth to eleventh aspects, wherein the determining further includes calculating a fourth evaluation value for each of the plurality of incidents, in the calculating of the fourth evaluation value, for each of the plurality of incidents, at least two fifth evaluation values are determined according to at least two values among: an expectation based on the distribution; a maximum value of the plurality of index values; a variance in the distribution; a proportion that is calculated based on the distribution and is accounted for by a distribution of index values greater than or equal to a first threshold value in the index; and an evaluation value determined using a machine learning model based on a plurality of incident trends corresponding to a plurality of first incident information items that are issued during a third period in the past, and the fourth evaluation value is calculated by adding up the at least two fifth evaluation values, and the priority is determined to be higher with an increase in the fourth evaluation value.

(61) An information notification method according to a twentieth aspect of the present disclosure is

the information notification method according to any one of the first to eighth aspects, wherein the determining includes determining, for each type of the plurality of incidents and for each model of monitoring targets: an incident trend of incidents of the type and the model during the second period in the future, the monitoring targets each being the monitoring target; and a priority of a provisional response or a permanent response to an incident of the type and the model.

(62) Therefore, it is possible to determine a priority of a provisional response or a permanent response for each incident type and for each monitoring target model.

(63) An information notification method according to a twenty first aspect of the present disclosure is the information notification method according to any one of the first to eighth aspects, wherein the determining further includes calculating, for each type of the plurality of incidents, a total amount of damage by multiplying a total occurrence count of incident information items of the type by an amount of damage that occurs when an incident information item of the type is issued, and the priority is determined to be higher with an increase in the total amount of damage.

(64) Therefore, it is possible to determine a priority based on a total amount of damage.

(65) An information notification method according to a twenty second aspect of the present disclosure is the information notification method according to any one of the first to eighth aspects, further including: calculating, for each type of the plurality of incidents, an index value about an occurrence count of incidents of the type during the second period, based on the incident trend determined for each type of the plurality of incidents; and notifying of recommendation of the provisional response when the index value is greater than a reference value.

(66) Therefore, it is possible to notify of recommendation of a provisional response when an index value about an occurrence count during the second period is greater than the reference value.

(67) An information notification method according to a twenty third aspect of the present disclosure is the information notification method according to the tenth aspect, further including: calculating a sixth evaluation value for each type of the plurality of incidents based on the incident trend determined for each type of the plurality of incidents; and notifying of recommendation of the provisional response when the sixth evaluation value is greater than a reference value.

(68) Therefore, it is possible to notify of recommendation of a provisional response when a sixth evaluation value is greater than the reference value.

(69) An information notification method according to a twenty fourth aspect of the present disclosure is the information notification method according to any one of the first to twenty second aspects, wherein the incident trend is determined using a machine learning model generated by machine learning using, as an input, a plurality of first incident information items that are issued during a third period in the past and using, as ground truth data, a track record of an incident trend based on a plurality of second incident information items that are issued during a fourth period in the past, the fourth period being later than the third period.

(70) Therefore, it is possible to predict a future incident trend of incident information by using a machine learning model.

(71) An information notification method according to a twenty fifth aspect of the present disclosure is the information notification method according to any one of the first to twenty second aspects, further including: determining a total amount of damage caused by a cyber attack during the second period by using a machine learning model generated by machine learning using, as an input, a plurality of first incident information items that are issued during a third period in the past and using, as ground truth data, a total amount of damage based on a plurality of second incident information items that are issued during a fourth period in the past, the fourth period being later than the third period.

(72) Therefore, it is possible to predict a total amount of damage caused by a future cyber attack by using a machine learning model.

(73) An information notification method according to a twenty sixth aspect of the present disclosure is the information notification method according to any one of the first to twenty second

aspects, wherein the incident trend is determined using a correlation between a plurality of first incident information items issued during a third period in the past and a track record of an incident trend based on a plurality of second incident information items issued during a fourth period in the past, the fourth period being later than the third period.

(74) Therefore, it is possible to predict an incident trend based on a past track record.

(75) An information notification method according to a twenty seventh aspect of the present disclosure is the information notification method according to any one of the first to twenty second aspects, further including: determining a total amount of damage caused by a cyber attack during the second period by using a correlation between a plurality of first incident information items issued during a third period in the past and a track record of a total amount of damage based on a plurality of second incident information items issued during a fourth period in the past, the fourth period being later than the third period.

(76) Therefore, it is possible to predict a total amount of damage caused by a future cyber attack based on a past track record.

(77) An information notification device according to an twenty eighth aspect of the present disclosure includes: a processor; and memory connected to the processor, wherein, by using the memory, the processor operates as: an obtainer that obtains a plurality of incident information items on a plurality of incidents caused by cyber attacks on a monitoring target during a first period in past; a determiner that determines, based on the plurality of incident information items obtained, for each type of the plurality of incidents, an incident trend of incidents of the type in the monitoring target during a second period in future and a priority of a provisional response or a permanent response to an incident of the type; and a presenter that presents the incident trend and the priority that are determined for each type of the plurality of incidents.

(78) Therefore, it is possible to provide information for decision-making to make it accurate and easy to determine whether to carry out the provisional response and determination of a response priority of the permanent response. For example, when a SIRT receives a notification, the SIRT can execute the provisional response or the permanent response in an early stage and can easily determine whether to carry out the provisional response and a priority of carrying out the permanent response. Accordingly, it is possible to preferentially respond to an incident that will have a great impact on security even with a limited number of members.

(79) It should be noted that general or specific aspects of the present disclosure may be realized as a system, a method, an integrated circuit, a computer program, a non-transitory computer-readable recording medium such as a CD-ROM, or any given combination thereof.

(80) Hereinafter, an information notification method and an information notification device according to an aspect of the present disclosure is described in detail with reference to the Drawings. It should be noted that the embodiment described below shows a specific example of the present disclosure. The numerical values, shapes, materials, constituent elements, the arrangement and connection of the constituent elements, steps, the order of the steps, etc., in the following embodiment are mere examples, and therefore do not intend to limit the present disclosure. Moreover, among the constituent elements in the following embodiment, constituent elements not recited in any one of the independent claims which indicate the broadest concepts of the present disclosure are described as arbitrary constituent elements.

Embodiment

(81) Hereinafter, an embodiment is described with reference to FIG. 1 to FIG. 10.

(82) [Configuration]

(83) FIG. 1 is a schematic view of an information notification system according to the embodiment.

(84) More specifically, FIG. 1 illustrates security monitoring device **100**, information notification device **200**, vehicle **400**, communication network **300**, and base station **310** of a mobile communication network. Security monitoring device **100**, information notification device **200**, and vehicle **400** are communicatively connected to send and receive information between each other

via communication network **300**.

(85) Security monitoring device **100** is a device that monitors a state of vehicle **400**. For example, security monitoring device **100** may have functions of a monitoring center (security operation center (SOC)) and SIRT. Security monitoring device **100** obtains, as a function of the monitoring center, log information periodically from vehicle **400** and monitors the state of vehicle **400** based on the obtained log information. More specifically, security monitoring device **100** determines the presence or absence of a cyber attack on vehicle **400**, based on the log information. By reducing time to next obtainment of the log information from vehicle **400**, security monitoring device **100** can determine the presence or absence of a cyber attack on vehicle **400** substantially in real time. When determining the presence of a cyber attack on vehicle **400**, security monitoring device **100** transmits incident information obtained from the determination to information notification device **200**.

(86) In addition, security monitoring device **100** obtains, as a function of the SIRT, notification information that indicates a predicted incident trend and a priority from information notification device **200** and presents the notification information to a member of the SIRT. Security monitoring device **100** accepts an input of response information that is generated through development referring to notification information from a member of a SIRT and transmits the response information to vehicle **400**.

(87) Note that the function of the monitoring center and the function of the SIRT that security monitoring device **100** has may be implemented by separate devices. To be more specific, in this case, a first device having the function of the monitoring center obtains the log information from vehicle **400** and outputs the generated incident information to information notification device **200**. A second device having the function of the SIRT obtains the notification information from information notification device **200** and outputs the generated response information to vehicle **400**.

(88) Note that although one vehicle **400** is illustrated in FIG. 1, information notification system **1** may include a plurality of vehicles **400**. More specifically, security monitoring device **100** may obtain log information periodically from each of the plurality of vehicles **400**, determine the presence or absence of a cyber attack for each of the plurality of vehicles **400**, and transmit incident information obtained for each of the plurality of vehicles **400** to information notification device **200**. Security monitoring device **100** generates the incident information when determining the presence of a cyber attack on vehicle **400**, and does not generate the incident information when determining the absence of a cyber attack on vehicle **400**. Security monitoring device **100** is implemented by a computer, such as a server.

(89) Information notification device **200** is a device that determines, based on a plurality of incident information items in the past, for each type of the plurality of incident information items, a future incident trend of incident information items of the type and a priority of a provisional response or a permanent response to an incident information item of the type and notifies of the determined incident trend and the priority. Information notification device **200** obtains a plurality of incident information items from security monitoring device **100** and, based on the plurality of incident information items obtained, estimates, for each type of the plurality of incident information items, a future incident trend of incident information items of the type. Information notification device **200** is implemented by a computer, such as a server.

(90) Vehicle **400** is an automated vehicle that is capable of autonomous driving. Vehicle **400** includes a presenter that presents information. Vehicle **400** controls an operation of vehicle **400** according to a received instruction. More specifically, according to the instruction, vehicle **400** may cause the presenter to present the information or control an operation related to motion of vehicle **400** (hereinafter, will be referred to as a motion operation). For example, vehicle **400** may be a vehicle used in a carsharing service or a vehicle used in a taxi service. Vehicle **400** is not limited to an automated vehicle capable of autonomous driving and may be an automobile that has a driver assistance function such as an ADAS function.

(91) Note that, in the present embodiment, security monitoring device **100** and information notification device **200** are separated. However, this should not be construed as limiting. Security monitoring device **100** and information notification device **200** may be integrated together. That is, security monitoring device **100** may have the functions of information notification device **200**.

(92) FIG. 2 is a block diagram illustrating an example of a hardware configuration of the information notification device according to the embodiment.

(93) As illustrated in FIG. 2, information notification device **200** has, as the hardware configuration, central processing unit (CPU) **21**, main memory **22**, storage **23**, and communication interface (communication IF) **24**.

(94) CPU **21** is a processor that executes a program stored in storage **23** or the like.

(95) Main memory **22** is a volatile storage area used as a work area when CPU **21** executes the program.

(96) Storage **23** is a nonvolatile storage area that holds the program, incident information, and the like.

(97) Communication IF **24** is a communication interface that communicates with security monitoring device **100** via communication network **300**. Communication IF **24** is, for example, a wired LAN interface. Note that communication IF **24** may be a wireless LAN interface. Furthermore, communication IF **24** is not limited to a LAN interface and may be any communication interface that is capable of establishing a communication connection with the communication network.

(98) FIG. 3 is a block diagram illustrating an example of a hardware configuration of the vehicle according to the embodiment.

(99) As illustrated in FIG. 3, vehicle **400** has, as a hardware configuration, telematics control unit (TCU) **41**, a plurality of ECUs **42**, and storage **43**.

(100) TCU **41** is a communication unit that enables vehicle **400** to perform wireless communication with communication network **300**. TCU **41** is a communication unit including a cellular module that meets a mobile communication network standard.

(101) The plurality of ECUs **42** are control circuits that control pieces of equipment included in vehicle **400**. Examples of the pieces of equipment included in vehicle **400** include an engine, a motor, a meter, a transmission, brakes, steering, power windows, and an air conditioner. Furthermore, at least one of the plurality of ECUs **42** is a control circuit that controls the autonomous driving of vehicle **400**. The plurality of ECUs **42** may be provided corresponding to these various pieces of equipment. Although not illustrated here, each of the plurality of ECUs **42** may include a storage (a nonvolatile storage area) that stores a program to be executed by ECU **42**. The storage is, for example, a nonvolatile memory.

(102) Storage **43** is a nonvolatile storage area that holds a control program and the like. Storage **43** is implemented by, for example, a hard disk drive (HDD), a solid stated drive (SSD), or the like.

(103) Next, functional configurations of information notification device **200** and vehicle **400** of information notification system **1** will be described. FIG. 4 is a block diagram illustrating an example of a functional configuration of the information notification system according to the embodiment. Note that communication network **300** is omitted in FIG. 4.

(104) First, the functional configuration of information notification device **200** will be described.

(105) Information notification device **200** includes communicator **210**, obtainer **220**, determiner **230**, notifier **240**, and storage **250**.

(106) Communicator **210** sends and receives information to and from security monitoring device **100** via communication network **300**. More specifically, communicator **210** receives incident information from security monitoring device **100**. The incident information received by communicator **210** is accumulated in storage **250**. In addition, communicator **210** transmits notification information to security monitoring device **100**. Note that communicator **210** is implemented by communication IF **24**.

(107) FIG. 5 is a table showing a specific example of incident statistical information.

(108) The incident statistical information is information generated based on incident information items that are caused by cyber attacks on vehicle **400** as a monitoring target during a past first period. As illustrated in FIG. 5, the incident statistical information, which is accumulated in storage **250**, includes ID, date, incident type, and occurrence count. The incident statistical information is statistical information about incident information that is generated based on a plurality of incident information items accumulated. The ID is information for identifying, for example, incidents of the same type that occur on the same day. The date indicates a date on which an incident occurs. The incident type indicates a type of an incident that occurs. For example, incident type is indicated by numbers. The occurrence count indicates a count of incidents of the same type that occur on the same day.

(109) FIG. 6 is a table showing another example of the information included in the incident statistical information.

(110) In FIG. 5, it is assumed that the incident statistical information includes a date on which an incident occurs as the information about an occurrence time point of the incident. Instead, as illustrated in FIG. 6, the incident statistical information may include a date and time at which an incident occurs.

(111) FIG. 7 is a table showing another example of the information included in the incident statistical information.

(112) The incident type in FIG. 5 is assumed to be indicated by a number indicating an incident type. However, this should not be construed as limiting. As illustrated in FIG. 7, the incident type may be indicated by information that represents characteristics of an incident type such as a method of attack and its severity or a feasibility of attack.

(113) FIG. 8 is a table showing another example of the information included in the incident statistical information.

(114) In FIG. 5, it is assumed that the incident statistical information includes an occurrence count that indicates a count of incidents of the same type that occur on the same day. However, this should not be construed as limiting. As illustrated in FIG. 8, the incident statistical information may include an occurrence frequency. For example, the occurrence frequency is defined in a graded manner according to counts of occurrences per day. The occurrence frequency may be classified into frequencies in five levels: “extremely low”, “low”, “middle”, “high”, and “extremely high”. The number of levels of frequencies is not limited to five. The number may be two to four or may be six or more.

(115) Note that vehicle **400** targeted for the cyber attacks is an example of a monitoring target. Note that the monitoring target is not limited to vehicle **400**. The monitoring target may be any device equipped with a computer and having a function that enables communication with an external network. That is, the application of the information notification system is not limited to monitoring a vehicle. The information notification system may be applied to IoT monitoring, server monitoring, building monitoring, factory monitoring, monitoring of a movable body such as an aircraft or a watercraft, or the like.

(116) Return to the description using FIG. 4.

(117) Obtainer **220** obtains, from storage **250**, the plurality of incident information items that are caused by cyber attacks on vehicle **400** during the past first period. Note that obtainer **220** is not limited to obtaining, from storage **250**, the plurality of incident information items that are issued during the first period. Obtainer **220** may obtain the plurality of incident information items from security monitoring device **100** and may obtain the incident statistical information.

(118) Based on the plurality of incident information items obtained or the incident statistical information, determiner **230** determines, for each incident information type, an incident trend of incident information items of the type in vehicle **400** during a future second period and a priority of a provisional response or a permanent response to an incident information item of the type. More

specifically, determiner **230** first determines the incident trend of the incident information of the type during the future second period (hereinafter, referred to as a “predicted incident trend”) and then determines the priority of the provisional response or the permanent response to the incident information of the type. The predicted incident trend relates to change over time in occurrence count of incident information items for each type.

(119) FIG. **9** is a graph illustrating change over time in occurrence count of incident information items.

(120) For example, the predicted incident trend is a slope of the change over time in occurrence count of incident information items of each type at first time point **t1** in the second period. That is, the predicted incident trend is a slope of a tangential line of the graph of the change over time at first time point **t1**. Alternatively, the predicted incident trend may be, for example, a curve (graph) illustrating change over time in occurrence count of incidents during the second period.

Alternatively, the predicted incident trend may be, for example, a total count of occurrence counts of incident information items during the second period. That is, the predicted incident trend may be the area between the curve illustrating the occurrence count during the second period and a line indicating that the occurrence count is zero in FIG. **9** (i.e., the area of the hatched zone in FIG. **9**).

(121) The predicted incident trend may be determined by using, for example, machine learning. In this case, the predicted incident trend is determined using a machine learning model generated by machine learning using, as an input, a plurality of first incident information items that are issued during a past third period and using, as ground truth data, a track record of an incident trend based on a plurality of second incident information items that are issued during a past fourth period later than the third period.

(122) Alternatively, the predicted incident trend may be determined by using, for example, a rule-based model. In this case, the predicted incident trend is determined using a correlation between the plurality of first incident information items issued during the past third period and the track record of the incident trend based on the plurality of second incident information items issued during the past fourth period later than the third period.

(123) After determining the predicted incident trend, determiner **230** determines, based on the incident trend of the incident information of each type, the priority of each type to be higher with an increase in potential future damage caused by a cyber attack. A higher priority indicates that the corresponding provisional response or the permanent response is made earlier. The priority may be determined to be higher with an increase in the slope as the predicted incident trend. The priority may be determined to be higher with an increase in the maximum value of the curve as the predicted incident trend. The priority may be determined to be higher with an increase in a time during which the curve as the predicted incident trend is greater than a given occurrence count. The priority may be determined to be higher with an increase in the total count of occurrence counts of the incident information items during the second period as the predicted incident trend.

(124) Return to the description using FIG. **4**.

(125) Notifier **240** notifies of the predicted incident trend and the priority that are determined for each type of incident information. More specifically, notifier **240** transmits notification information indicating the predicted incident trend and the priority to security monitoring device **100** via communicator **210**.

(126) Obtainer **220**, determiner **230**, and notifier **240** are implemented by CPU **21**, main memory **22**, and the like. Storage **250** is implemented by storage **23**.

(127) [Operation]

(128) FIG. **10** is a sequence diagram illustrating an example of an information notification method executed by the information notification system according to the embodiment.

(129) Vehicle **400** transmits the log information to security monitoring device **100** (**S11**). For example, the log information indicates a control status of vehicle **400**, a detection value of a sensor included in vehicle **400**, or the like.

(130) Security monitoring device **100** detects the presence of a cyber attack on vehicle **400** based on the log information (S12).

(131) Based on the cyber attack, security monitoring device **100** generates the above-mentioned incident information and transmits the incident information to information notification device **200** (S13).

(132) Information notification device **200** receives the incident information (S14).

(133) Information notification device **200** stores the received incident information in storage **250** (S15).

(134) Information notification device **200** loops step S16 and step S17 for each type of incident.

(135) Information notification device **200** reads incident information of a type targeted for processing from storage **250** (S16). The incident information read here includes a plurality of incident information items generated at cyber attacks that occur during the past first period.

(136) Based on the plurality of incident information items generated at the cyber attacks occurring during the past first period, information notification device **200** determines the predicted incident trend (S17).

(137) Based on the predicted incident trend, information notification device **200** determines the priority of each type of incident (S18). Information notification device **200** notifies security monitoring device **100** of the predicted incident trend and the priority (S19).

(138) Referring to information on the obtained predicted incident trend and the priority, security monitoring device **100** determines whether to carry out a provisional response and details of the response or gives a priority of a permanent response for each incident type. Security monitoring device **100** accepts an input of response information that is used for causing vehicle **400** to execute a provisional response or a permanent response to respond to an incident for each type and generated by a member of the SIRT, and transmits the accepted response information to vehicle **400** (S20).

(139) Note that the priority may be relative between a plurality of incident types or may be an absolute one calculated by comparison with a predetermined reference.

(140) The provisional response is, for example, a response to stop a function involving a vulnerability that results in a cause of a cyber attack or to stop a function that will be affected by the vulnerability. Note that, in a case where there are a plurality of methods for a provisional response responding to a target incident type, a method for the provisional response may be determined using a priority of each type of incident.

(141) The permanent response is, for example, a response to execute a software update to overcome the vulnerability that results in the cause of the cyber attack.

(142) [Advantageous Effects Etc.]

(143) The information notification method according to the present embodiment includes: obtaining the plurality of incident information items caused by cyber attacks on vehicle **400** (the monitoring target) during the past first period; based on the plurality of incident information items obtained, determining, for each type of the plurality of incident information items, the incident trend (the predicted incident trend) of incident information items of the type in vehicle **400** (the monitoring target) during the future second period and the priority of the provisional response or the permanent response to an incident information item of the type; and notifying of the predicted incident trend and the priority that are determined for each type of the plurality of incident information items.

(144) Therefore, it is possible to provide information for decision-making to make it accurate and easy to determine whether to carry out the provisional response and determination of a response priority of the permanent response. For example, when the SIRT receives the notification, the SIRT can easily determine whether to carry out the provisional response and the priority of carrying out the permanent response, thus executing the provisional response or the permanent response in an early stage. Accordingly, it is possible to preferentially respond to an incident that will have a great

impact on security even with a limited number of members.

(145) In addition, in the information notification method according to the present embodiment, a higher priority indicates that the provisional response or the permanent response is made earlier, and based on the incident trend, the priority is determined to be higher with an increase in potential future damage caused by a cyber attack. Accordingly, it is possible to preferentially respond to an incident in which a cyber attack will cause great potential future damage even with a limited number of members.

(146) In addition, in the information notification method according to the present embodiment, the predicted incident trend relates to the change over time in occurrence count of the incident information items for each type. Therefore, for example, when the SOC or the SIRT receives the notification, the SOC or the SIRT can easily grasp a future incident trend of incident information. Accordingly, it is possible to perform efficient monitoring and response through changing a personnel distribution or the like.

(147) In addition, in the information notification method according to the present embodiment, the predicted incident trend is the slope of the change over time at the first time point in the second period. In addition, the priority is determined to be higher with an increase in the slope. Accordingly, the priority of the response is determined to be higher for an incident that has a steeper slope and has an increasing tendency. Thus, it is possible to promote a more prioritized response for an incident that has a steeper slope and has an increasing tendency.

(148) In addition, in the information notification method according to the present embodiment, the predicted incident trend is the curve illustrating the change over time during the second period. In addition, the priority is determined to be higher with an increase in the maximum value of the curve. Accordingly, the priority of the response is determined to be higher for an incident that has a curve with a larger maximum value. Thus, it is possible to promote a more prioritized response for an incident that has a curve with a larger maximum value.

(149) In addition, in the information notification method according to the present embodiment, the priority is determined to be higher with an increase in the time during which the curve is greater than the given occurrence count. Accordingly, a priority of a response for an incident having a longer time during which its curve is greater than or equal to the given occurrence count is determined to be higher. Thus, it is possible to promote a more prioritized response for an incident having a longer time during which its curve is greater than or equal to the given occurrence count.

(150) In addition, in the information notification method according to the present embodiment, the predicted incident trend is the total count of occurrence counts of the incident information items during the second period. In addition, the priority is determined to be higher with an increase in the total count. Accordingly, the priority of the response is determined to be higher for an incident that causes a higher total count of occurrence counts during the second period. Thus, it is possible to promote a more prioritized response for an incident that causes a higher total count of occurrence counts during the second period.

(151) In addition, in the information notification method according to the present embodiment, the predicted incident trend is determined using the machine learning model generated by machine learning using, as an input, the plurality of first incident information items that are issued during the past third period and using, as ground truth data, the track record of the incident trend based on the plurality of second incident information items that are issued during the past fourth period later than the third period. Therefore, it is possible to predict the future incident trend of incident information by using the machine learning model.

(152) In addition, in the information notification method according to the present embodiment, the predicted incident trend is determined using the correlation between the plurality of first incident information items issued during the past third period and the track record of the incident trend based on the plurality of second incident information items issued during the past fourth period later than the third period. Therefore, it is possible to predict the incident trend based on the past

track record.

(153) Variation

(154) (1) Variation 1

(155) In the above-described embodiment, the reading of the incident information (S16) and the determination of the predicted incident trend (S17) are performed for each type of incident, and the priority is determined for each type of incident. However, this should not be construed as limiting.

(156) FIG. 11 is a sequence diagram illustrating an example of an information notification method executed by an information notification system according to Variation 1. Note that, in FIG. 11, the same processing as in the embodiment will be denoted by the same reference sign as in FIG. 10, and the description thereof will be omitted.

(157) For example, as illustrated in FIG. 11, further for each model of vehicles 400 (i.e., monitoring targets), the reading of the incident information (S16) and the determination of the predicted incident trend (S17) may be performed for each type of incident, and the priority may be determined for each type of incident. That is, the predicted incident trend and the priority may be determined for each model and for each type of incident. Likewise, the predicted incident trend and the priority may be determined for each type of incident and for each model.

(158) (2) Variation 2

(159) In the above-described embodiment, the priority is determined based on the predicted incident trend. However, the priority may be determined based on a total amount of damage that is further calculated from the predicted incident trend.

(160) FIG. 12 is a sequence diagram illustrating an example of an information notification method executed by an information notification system according to Variation 2. Note that, in FIG. 12, the same processing as in the embodiment will be denoted the same reference sign as in FIG. 10, and the description thereof will be omitted.

(161) After step S17, information notification device 200 determines a total amount of damage for each type of incident. For example, information notification device 200 calculates the total amount of damage (a total damage severity) by multiplying a total occurrence count of incident information items of the type by an amount of damage (a damage severity) that occurs when an incident information item of the type is issued.

(162) Alternatively, the total amount of damage may be calculated using a machine learning model generated by machine learning using, as an input, the plurality of first incident information items that are issued during the past third period and using, as ground truth data, a track record of a total amount of damage based on the plurality of second incident information items that are issued during the past fourth period later than the third period.

(163) Alternatively, the total amount of damage may be determined using the correlation between the plurality of first incident information items issued during the past third period and the track record of the total amount of damage based on the plurality of second incident information items issued during the past fourth period later than the third period.

(164) Information notification device 200 determines the priority to be higher with an increase in the total amount of damage (total damage severity) (S18a).

(165) Information notification device 200 notifies security monitoring device 100 of the predicted incident trend, the priority, and the total damage severity (S19a).

(166) Accordingly, the priority of the response is determined to be higher for an incident that will result in a larger total amount of damage. Thus, it is possible to promote a more prioritized response for an incident that will result in a larger total amount of damage.

(167) (3) Variation 3

(168) FIG. 13 is a sequence diagram illustrating an example of an information notification method executed by an information notification system according to Variation 3. Note that, in FIG. 13, the same processing as in the embodiment will be denoted the same reference sign as in FIG. 10, and the description thereof will be omitted.

(169) After step **S17**, information notification device **200** determines whether a future occurrence count is greater than a reference value (**S17b**).

(170) When the future occurrence count is greater than the reference value (Yes in **S17b**), information notification device **200** notifies security monitoring device **100** of recommendation to carry out the provisional response (**S17c**). Otherwise (No in **S17b**), information notification device **200** performs the next processing. It is not limited to determining whether the future occurrence count is greater than the reference value. It may be determined whether an occurrence probability obtained by dividing an occurrence count of incidents by the number of vehicles of each model is greater than a threshold value. In this case, when the occurrence probability is greater than the threshold value, step **S17c** is performed. Otherwise, the next processing is performed. It is not limited to determining whether the future occurrence count is greater than the reference value. The total amount of damage may be calculated for each type of incident, and it may be determined whether a future total amount of damage is greater than a threshold value may be determined. In this case, when the total amount of damage is greater than the threshold value, step **S17c** is performed. Otherwise, the next processing is performed.

(171) As seen from the above, the incident trend is determined for each type of incident, and based on the incident trend of the type of incident, information notification device **200** calculates an index value about an occurrence count of the type of incident during the second period. The index value is, as mentioned above, the future occurrence count, the future occurrence probability, the future total amount of damage, or the like.

(172) (4) Variation **4**

(173) In Variation **4**, an incident trend of incident information of one type may include a distribution of a plurality of index values that are likely to be determined in an index indicating the incident trend. More specifically, the index indicating the incident trend is any one of the followings: (i) a slope of the change over time in occurrence count of incident information items of each type at first time point **t1** in the second period, (ii) a maximum value of a curve (graph) illustrating change over time in occurrence count of incidents during the second period, (iii) a time during which the curve (graph) illustrating the change over time in occurrence count of incidents during the second period is greater than the given occurrence count, and (iv) the total count of occurrence counts of incident information items during the second period.

(174) As illustrated in FIG. **14**, the distribution of the plurality of index values may be given by a probability density function with a random variable that is an index indicating an incident trend corresponding to the distribution. FIG. **14** is a graph illustrating an example of the probability density function with the random variable that is the index indicating the incident trend. In the probability density function, its horizontal axis represents the incident trend, and its vertical axis represents the occurrence probability of the corresponding incident trend. For example, in a case where the incident trend is the slope of the change over time in occurrence count of incident information items of each type at first time point **t1** in the second period, the probability density function represents the relation between a slope and the density of the occurrence probability of the slope.

(175) Alternatively, as illustrated in FIG. **15**, the distribution of the plurality of index values may be given by a range including a plurality of index values that can take on in the index indicating the incident trend corresponding to the distribution. FIG. **15** is a diagram illustrating an example in which the distribution is given by the range including the plurality of index values that can take on in the index indicating the incident trend. A plurality of points in FIG. **15** illustrate the plurality of index values that can take on in the index indicating the incident trend. The range including the plurality of index values is represented by, for example, a range between a minimum value and a maximum value of the plurality of index values.

(176) Alternatively, as illustrated in FIG. **16**, the distribution of the plurality of index values may be the distribution of the plurality of index values itself. FIG. **16** is a diagram illustrating an example

in which the distribution is given by the plurality of index values that can take on in the index indicating the incident trend. A plurality of points in FIG. 16 illustrate the plurality of index values that can take on in the index indicating the incident trend.

(177) Next, a priority determination method in a case where the distribution of the plurality of index values is given by the probability density function will be described.

(178) FIG. 17 is a diagram for describing a first example of the priority determination method in Variation 4. FIG. 17 illustrates expectation E1 of the incident trend in probability density function f1 of incident information of type No. 1, expectation E2 of the incident trend in probability density function f2 of incident information of type No. 2, and expectation E3 of the incident trend in probability density function f3 of incident information of type No. 3. In the first example, the priority may be determined to be higher with an increase in the expectation based on the probability density function.

(179) In a case of the example in FIG. 17, expectation E2 is larger than expectation E1, and expectation E1 is larger than expectation E3. Therefore, a priority of the incident information of No. 2 may be determined to be higher than a priority of the incident information of No. 1, and the priority of the incident information of No. 1 may be determined to be higher than a priority of the incident information of No. 3. Note that the incident information of No. 1, the incident information of No. 2, and the incident information of No. 3 are different in type from one another. The priority of incident information indicates a priority of the provisional response or the permanent response to the incident information.

(180) FIG. 18 is a diagram for describing a second example of the priority determination method in Variation 4. FIG. 18 illustrates maximum value MAX1 of a domain of the incident trend in probability density function f1 of the incident information of type No. 1, maximum value MAX2 of a domain of the incident trend in probability density function f2 of the incident information of type No. 2, and maximum value MAX3 of a domain of the incident trend in probability density function f3 of incident information of type No. 3. In the second example, the priority may be determined to be higher with an increase in the maximum value of the domain of the incident trend in the probability density function.

(181) In a case of the example in FIG. 18, maximum value MAX1 is larger than maximum value MAX2, and maximum value MAX2 is larger than maximum value MAX3. Therefore, the priority of the incident information of No. 1 may be determined to be higher than the priority of the incident information of No. 2, and the priority of the incident information of No. 2 may be determined to be higher than the priority of the incident information of No. 3.

(182) FIG. 19 is a diagram for describing a third example of the priority determination method in Variation 4. FIG. 19 illustrates variance V1 of the incident trend in probability density function f1 of the incident information of type No. 1, variance V2 of the incident trend in probability density function f2 of the incident information of type No. 2, and variance V3 of the incident trend in probability density function f3 of incident information of type No. 3. Note that FIG. 19 is not a precise illustration of variance V1, variance V2, and variance V3 but a schematic view. In the third example, the priority may be determined to be higher with an increase in the variance based on the probability density function.

(183) In a case of the example in FIG. 19, variance V3 is larger than variance V1, and variance V1 is larger than variance V2. Therefore, the priority of the incident information of No. 3 may be determined to be higher than the priority of the incident information of No. 1, and the priority of the incident information of No. 1 may be determined to be higher than a priority of the incident information of No. 2. As seen from the above, by determining the priority to be higher with an increase in variance, it is possible to determine the priority to be higher with an increase in variation in the incident trend. That is, it is possible to determine the priority to be higher with an increase in risk incurred by the incident trend.

(184) FIG. 20 is a diagram for describing a fourth example of the priority determination method in

Variation 4. FIG. 20 illustrates an occurrence probability (e.g., 15%) that the incident trend in probability density function f_1 of the incident information of No. 1 is greater than or equal to threshold value Th_1 , an occurrence probability (e.g., 30%) that the incident trend in probability density function f_2 of the incident information of No. 2 is greater than or equal to threshold value Th_1 , and an occurrence probability (e.g., 5%) that the incident trend in probability density function f_3 of incident information of No. 3 is greater than or equal to threshold value Th_1 . In the fourth example, the priority may be determined to be higher with an increase in the occurrence probability that the random variable of the probability density function is greater than or equal to threshold value Th_1 .

(185) FIG. 21 is a diagram for describing a fifth example of the priority determination method in Variation 4. The fifth example is an example of a priority determination method using a machine learning model. The machine learning model is generated by machine learning using, as an input, a plurality of incident trends corresponding to the plurality of first incident information items that are issued during the past third period and using, as ground truth data, an evaluation value determined for each of the plurality of incident trends. The evaluation values in the ground truth data are an example of first evaluation values. For example, the evaluation values in the ground truth data may be values determined by a person evaluating the incident trends or may be a score based on at least one of the expectation, maximum value, and variance used for calculating the priority determined based on the above-mentioned probability density function. By inputting an incident trend corresponding to incident information into the machine learning model, an evaluation value corresponding to the incident information is calculated. In the fifth example, the priority may be determined to be higher with an increase in the calculated evaluation value.

(186) FIG. 22 is a diagram for describing a sixth example of the priority determination method in Variation 4. The sixth example is another example of the priority determination method using a machine learning model. The machine learning model is generated by machine learning using, as an input, a plurality of incident trends corresponding to a plurality of first incident information items of another type that are issued during the past third period and using, as ground truth data, a precedence determined for each of the plurality of incident trends. For example, the precedence in the ground truth data may be precedence determined by a person for the plurality of incident trends or may be precedence determined based on the priority determined based on the above-mentioned probability density function. In the latter case, the precedence is determined to be higher with an increase in the priority. By inputting, into the machine learning model, a plurality of incident trends corresponding to a plurality of incident information items that are different in type from one another, a precedence corresponding to each of the plurality of incident information items is determined. The higher a precedence of a type of incident information, the more preferentially the provisional response or the permanent response to the type of incident information is made.

(187) FIG. 23 is a diagram for describing a seventh example of the priority determination method in Variation 4. In the seventh example, evaluation values corresponding to values of the expectation, the maximum value, and the variance determined in the first example to the third example are calculated. The evaluation values are an example of second evaluation values. For example, as illustrated in FIG. 23, for the incident information of No. 1, an evaluation value of expectation E_1 is calculated to be 2 points because expectation E_1 is the second largest value of expectations E_1 to E_3 , an evaluation value of maximum value MAX_1 is calculated to be 3 points because maximum value MAX_1 is the largest value of maximum values MAX_1 to MAX_3 , and an evaluation value of variance V_1 is calculated to be 2 points because variance V_1 is the second largest value of variances V_1 to V_3 . Likewise, evaluation values of expectation E_2 , maximum value MAX_2 , and variance V_2 of the incident information of No. 2 are calculated, and evaluation values of expectation E_3 , maximum value MAX_3 , and variance V_3 of the incident information of No. 3 are calculated. As seen from the above, evaluation values for each of an expectation, a maximum value, and a variance are calculated by ranking a plurality of incident information items

according to a corresponding one of the expectation, the maximum value, and the variance. For example, an evaluation value is calculated to be higher as the incident information item is ranked higher.

(188) Then, the evaluation value of expectation, the evaluation value of maximum value, and the evaluation value of variance calculated for each type of incident information are added up, and thus a score is calculated for each type. The scores are an example of third evaluation values.

Accordingly, a score of the incident information of No. 1 is calculated to be 7 points, a score of the incident information of No. 2 is calculated to be 6 points, and a score of the incident information of No. 3 is calculated to be 5 points.

(189) In the seventh example, an example of calculating evaluation values corresponding to an expectation, a maximum value, and a variance determined in the first example to the third example has been described. However, evaluation values may be calculated based on at least two values among: an expectation; a maximum value; a variance; a proportion of a distribution of index values greater than or equal to threshold value Th1 calculated based on the distributions in the fourth example to the sixth example; and an evaluation value determined using a machine learning model.

(190) As seen from the above, when determining the priority, information notification device **200** calculates at least two second evaluation values for each of a plurality of incidents by ranking the plurality of incidents according to each of at least two values among: an expectation based on a probability density function; a maximum value of a domain of a random variable; a variance and proportion in the probability density function; and an evaluation value determined using a machine learning model. Information notification device **200** then calculates a third evaluation value for each of the plurality of incidents by adding up the at least two second evaluation values. In addition, the priority is here determined to be higher with an increase in the third evaluation value.

(191) FIG. **24** is a diagram for describing an eighth example of the priority determination method in Variation **4**. In the eighth example, evaluation values corresponding to values of the expectation, the maximum value, and the variance determined in the first example to the third example are calculated. The evaluation values are an example of fifth evaluation values. The values of expectation, maximum value, and variance are classified into a plurality of ranks in accordance with magnitudes of the values, and evaluation values corresponding to the classified ranks are calculated as evaluation values corresponding to the values. For example, in FIG. **24**, the values of expectation, maximum value, and variance are classified into three ranks: large, middle, and small in accordance with magnitudes of the values. For the incident information of No. 1, expectation E1 is classified as middle. Thus, an evaluation value of expectation E1 is calculated to be 2 points. Likewise, maximum value MAX1 is classified as large. Thus, an evaluation value of maximum value MAX1 is calculated to be 5 points. Likewise, variance V1 is classified as middle. Thus, an evaluation value of variance V1 is calculated to be 2 points.

(192) Then, the evaluation value of expectation E1, the evaluation value of maximum value MAX1, and the evaluation value of variance V1 calculated for the incident information of type No. 1 are added up, and thus a score is calculated. Accordingly, a score of the incident information of No. 1 is calculated to be 9 points. FIG. **24** illustrates a method of calculating the score of the incident information of type No. 1 as an example, and scores of incident information of other types are calculated likewise. The scores are an example of the fourth evaluation values.

(193) In the eighth example, an example of calculating evaluation values corresponding to an expectation, a maximum value, and a variance determined in the first example to the third example has been described. However, evaluation values may be calculated based on at least two values among: an expectation; a maximum value; a variance; a proportion of a distribution of index values greater than or equal to threshold value Th1 calculated based on the distributions in the fourth example to the sixth example; and an evaluation value determined using a machine learning model.

(194) As seen from the above, when determining the priority, information notification device **200** calculates a fourth evaluation value for each of a plurality of incidents. A fourth evaluation value is

calculated by (i) determining, for a corresponding incident, at least two fifth values according to at least two values among: an expectation based on a probability density function; a maximum value of a domain of a random variable in the probability density function; a variance in the probability density function; a proportion that is calculated based on the distribution and is accounted for by a distribution of index values greater than or equal to a first threshold value in the index; and an evaluation value determined using a machine learning model based on a plurality of incident trends corresponding to a plurality of first incident information items that are issued during the past third period, and (ii) adding up the at least two fifth evaluation values.

(195) The priority determination methods in the first to eighth examples are described as an example of a case where the incident trend is given by a probability density distribution function as illustrated in FIG. 14. Likewise, the methods may be applied to a case where the incident trend is given by a range including a plurality of index values as illustrated in FIG. 15 or a case where the incident trend is given by a plurality of index values themselves as illustrated in FIG. 16.

(196) For example, in a case where the priority determination method in the first example is applied to a distribution given by a range, a median of the entire range representing the distribution may be calculated as the expectation. Alternatively, in a case where the priority determination method in the first example is applied to a distribution given by a plurality of index values themselves, a mean of the plurality of index values may be calculated as the expectation. That is, in the first example, the priority is determined to be higher with an increase in an expectation based on a distribution of the plurality of index values that are likely to be determined in an index indicating the incident trend.

(197) For example, in a case where the priority determination method in the second example is applied to a distribution given by a range, a maximum value of the range representing the distribution may be calculated as the maximum value. Alternatively, in a case where the priority determination method in the second example is applied to a distribution given by a plurality of index values themselves, a maximum value of the plurality of index values may be calculated as the maximum value. That is, in the second example, the priority is determined to be higher with an increase in a maximum value of the plurality of index values that are likely to be determined in an index indicating the incident trend.

(198) For example, in a case where the priority determination method in the third example is applied to a distribution given by a range, a size (width) of the range representing the distribution may be calculated as the variance. That is, in the third example, the priority is determined to be higher with an increase in a variance based on a distribution of the plurality of index values that are likely to be determined in an index indicating the incident trend.

(199) For example, in a case where the priority determination method in the fourth example is applied to a distribution given by a range, a proportion of a range where the incident trend is greater than or equal to threshold value Th1 to the entire range representing the distribution may be calculated instead of an occurrence probability that the incident trend in the probability density function is greater than or equal to threshold value Th1. Alternatively, in a case where the priority determination method in the fourth example is applied to a distribution given by a plurality of index values themselves, the number of index values greater than or equal to threshold value Th1 out of the plurality of index values may be calculated instead of an occurrence probability that the incident trend in the probability density function is greater than or equal to threshold value Th1. As seen from the above, the priority may be determined to be higher with an increase in a proportion that is calculated based on a distribution of the plurality of index values that are likely to be determined in an index indicating the incident trend and is accounted for by a distribution of index values greater than or equal to threshold value Th1 in the index.

(200) FIG. 25 is a sequence diagram illustrating an example of an information notification method executed by an information notification system according to Variation 4. Note that, in FIG. 25, the same processing as in the embodiment will be denoted the same reference sign as in FIG. 10, and

the description thereof will be omitted.

(201) After step **S17**, information notification device **200** determines whether an occurrence probability that the future occurrence count is greater than or equal to reference θ is greater than a reference value (N %) (**S17d**). N is a value greater than 0 and less than 100.

(202) When an occurrence probability that the future occurrence count is greater than the reference value (N %) (Yes in **S17d**), information notification device **200** notifies security monitoring device **100** of recommendation to carry out the provisional response (**S17c**). Otherwise (No in **S17d**), information notification device **200** performs the next processing.

(203) As seen from the above, the incident trend is determined for each type of incident, and based on the incident trend of the type of incident, information notification device **200** calculates a sixth evaluation value. The sixth evaluation value is, for example, an occurrence probability that the future occurrence count is greater than or equal to reference θ . The sixth evaluation value is not limited to the occurrence probability as described above. The sixth evaluation value may be a value that has a correlation with the above-mentioned priority in a case where the priority is determined based on the probability density function (e.g., the expectation, the maximum value, the variance, or the score (the second evaluation value, the third evaluation value, or the fifth evaluation value)). Then, information notification device **200** may notify of recommendation of the provisional response when the sixth evaluation value is greater than a reference value (N %) and may skip the notification of the recommendation of the provisional response when the sixth evaluation value is less than or equal to the reference value.

(204) The priority may be indicated as not only a numerical value or a precedence but also a rank classified based on a numerical value. The priority may be indicated as a rank that is classified into a plurality of levels including, for example, a first rank corresponding to a first range, a second rank corresponding to a second range, and a third rank corresponding to a third range. The first rank is higher in priority than the second rank, and the second rank is higher in priority than the third rank. The first range is a range from a first value to a second value smaller than the first value, the second range is a range from a third value smaller than the second value to a fourth value smaller than the third value, and the third range is a range from a fifth value smaller than the fourth value to a sixth value smaller than the fifth value.

(205) (5) Variation 5

(206) Each of the elements in the embodiment described above may be configured in the form of an exclusive hardware product, or may be realized by executing a software program suitable for the element. Each of the elements may be realized by means of a program executing unit, such as a Central Processing Unit (CPU) or a processor, reading and executing the software program recorded on a recording medium such as a hard disk or semiconductor memory. Here, the software implementing information notification device **200** in the above-described embodiment is a program described below.

(207) In other words, this program causes a computer to execute the information notification method including: obtaining the plurality of incident information items caused by cyber attacks on the monitoring target during the past first period; based on the plurality of incident information items obtained, determining, for each type of the plurality of incident information items, the incident trend of incident information items of the type in the monitoring target during the future second period and the priority of the provisional response or the permanent response to an incident information item of the type; and notifying of the incident trend and the priority that are determined for each type of the plurality of incident information items.

(208) Herein above, although an information notification method according to one or more aspects of the present disclosure has been described based on the embodiment, the present disclosure is not limited to the above-described embodiment. Various modifications of the embodiment as well as other embodiments resulting from combinations of some of the constituent elements from the embodiment and variations that may be conceived by those skilled in the art may be included

within the scope of the present disclosure as long as they do not depart from the essence of the present disclosure.

(209) While various embodiments have been described herein above, it is to be appreciated that various changes in form and detail may be made without departing from the spirit and scope of the present disclosure as presently or hereafter claimed.

(210) Further Information about Technical Background to this Application

(211) The disclosures of the following patent applications including specification, drawings, and claims are incorporated herein by reference in their entirety: Japanese Patent Application No. 2022-202232 filed on Dec. 19, 2022, Japanese Patent Application No. 2023-206062 filed on Dec. 6, 2023, and PCT International Application No. PCT/JP2023/044743 filed on Dec. 13, 2023.

INDUSTRIAL APPLICABILITY

(212) The present disclosure is useful as an information notification method and the like capable of providing information for decision-making to make it accurate and easy to determine whether to carry out the provisional response and determination of a response priority of the permanent response.

Claims

1. An information notification method comprising: obtaining a plurality of incident information items on a plurality of incidents caused by cyber attacks on a monitoring target during a first period, wherein the first period is earlier than a current point in time, and wherein the monitoring target is a vehicle capable of movement; determining, based on the plurality of incident information items obtained, for each type of the plurality of incidents: an incident trend of incidents of the type in the monitoring target during a second period; and a priority of a provisional response which is a response to stop a function involving a vulnerability or to stop a function or a permanent response which is a response to execute a software update to overcome the vulnerability to an incident of the type, wherein the second period is later than the current point in time; presenting the incident trend and the priority that are determined for each type of the plurality of incidents and are used to perform an operation of the vehicle.
2. The information notification method according to claim 1, wherein the priority being a higher priority indicates that the provisional response or the permanent response is made earlier, and based on the incident trend, the priority is determined to be higher with an increase in potential future damage caused by a cyber attack.
3. The information notification method according to claim 2, wherein the incident trend relates to change over time in occurrence count of incidents for each type of the plurality of incidents.
4. The information notification method according to claim 3, wherein the incident trend is a slope of the change over time at a first time point in the second period, and the priority is determined to be higher with an increase in the slope.
5. The information notification method according to claim 3, wherein the incident trend is a curve illustrating the change over time during the second period, and the priority is determined to be higher with an increase in a maximum value of the curve.
6. The information notification method according to claim 3, wherein the incident trend is a curve illustrating the change over time during the second period, and the priority is determined to be higher with an increase in a time during which the curve is greater than a given occurrence count.
7. The information notification method according to claim 3, wherein the incident trend is a total count of occurrence counts of incidents during the second period, and the priority is determined to be higher with an increase in the total count.
8. The information notification method according to claim 3, wherein the incident trend is a total count of occurrence counts of incidents during the second period, and the priority is determined to be higher with an increase in a total amount of damage that is calculated by multiplying the total

count by an amount of damage that occurs when one incident occurs.

9. The information notification method according to claim 1, wherein the incident trend includes a distribution of a plurality of index values that are likely to be determined in an index indicating the incident trend in the determining.

10. The information notification method according to claim 9, wherein the distribution is given by a probability density function with a random variable that is the index indicating the incident trend.

11. The information notification method according to claim 9, wherein the distribution is given by a first range that includes the plurality of index values.

12. The information notification method according to claim 9, wherein the priority is determined to be higher with an increase in an expectation based on the distribution.

13. The information notification method according to claim 9, wherein the priority is determined to be higher with an increase in a maximum value of the plurality of index values.

14. The information notification method according to claim 9, wherein the priority is determined to be higher with an increase in a variance in the distribution.

15. The information notification method according to claim 9, wherein the priority is determined to be higher with an increase in a proportion that is calculated based on the distribution and is accounted for by a distribution of index values greater than or equal to a first threshold value in the index.

16. The information notification method according to claim 9, wherein the priority is determined using a machine learning model generated by machine learning using, as an input, a plurality of incident trends corresponding to a plurality of first incident information items that are issued during a third period and using, as ground truth data, a first evaluation value determined for each of the plurality of incident trends, wherein the third period is earlier than the current point in time.

17. The information notification method according to claim 9, wherein a precedence based on the priority is determined using a machine learning model generated by machine learning using, as an input, a plurality of incident trends corresponding to a plurality of first incident information items that are issued during a third period and using, as ground truth data, a precedence determined for each of the plurality of incident trends, wherein the third period is earlier than the current point in time.

18. The information notification method according to claim 9, wherein the determining further includes: calculating at least two second evaluation values for each of the plurality of incidents by ranking the plurality of incidents according to each of at least two values among: an expectation based on the distribution; a maximum value of the plurality of index values; a variance in the distribution; a proportion that is calculated based on the distribution and is accounted for by a distribution of index values greater than or equal to a first threshold value in the index; and an evaluation value determined using a machine learning model based on a plurality of incident trends corresponding to a plurality of first incident information items that are issued during a third period, wherein the third period is earlier than the current point in time; and calculating a third evaluation value for each of the plurality of incidents by adding up the at least two second evaluation values, and the priority is determined to be higher with an increase in the third evaluation value.

19. The information notification method according to claim 9, wherein the determining further includes calculating a fourth evaluation value for each of the plurality of incidents, in the calculating of the fourth evaluation value, for each of the plurality of incidents, at least two fifth evaluation values are determined according to at least two values among: an expectation based on the distribution; a maximum value of the plurality of index values; a variance in the distribution; a proportion that is calculated based on the distribution and is accounted for by a distribution of index values greater than or equal to a first threshold value in the index; and an evaluation value determined using a machine learning model based on a plurality of incident trends corresponding to a plurality of first incident information items that are issued during a third period, wherein the third period is earlier than the current point in time, and the fourth evaluation value is calculated by

adding up the at least two fifth evaluation values, and the priority is determined to be higher with an increase in the fourth evaluation value.

20. The information notification method according to claim 1, wherein the determining includes determining, for each type of the plurality of incidents and for each model of monitoring targets: an incident trend of incidents of the type and the model during the second period, the monitoring targets each being the monitoring target; and a priority of a provisional response or a permanent response to an incident of the type and the model.

21. The information notification method according to claim 1, wherein the determining further includes calculating, for each type of the plurality of incidents, a total amount of damage by multiplying a total occurrence count of incident information items of the type by an amount of damage that occurs when an incident information item of the type is issued, and the priority is determined to be higher with an increase in the total amount of damage.

22. The information notification method according to claim 1, further comprising: calculating, for each type of the plurality of incidents, an index value about an occurrence count of incidents of the type during the second period, based on the incident trend determined for each type of the plurality of incidents; and notifying of recommendation of the provisional response when the index value is greater than a reference value.

23. The information notification method according to claim 10, further comprising: calculating a sixth evaluation value for each type of the plurality of incidents based on the incident trend determined for each type of the plurality of incidents; and notifying of recommendation of the provisional response when the sixth evaluation value is greater than a reference value.

24. The information notification method according to claim 1, wherein the incident trend is determined using a machine learning model generated by machine learning using, as an input, a plurality of first incident information items that are issued during a third period and using, as ground truth data, a track record of an incident trend based on a plurality of second incident information items that are issued during a fourth period, wherein each of the third period and the fourth period is earlier than the current point in time, and the fourth period being later than the third period.

25. The information notification method according to claim 1, further comprising: determining a total amount of damage caused by a cyber attack during the second period by using a machine learning model generated by machine learning using, as an input, a plurality of first incident information items that are issued during a third period and using, as ground truth data, a total amount of damage based on a plurality of second incident information items that are issued during a fourth period, wherein each of the third period and the fourth period is earlier than the current point in time, and the fourth period being later than the third period.

26. The information notification method according to claim 1, wherein the incident trend is determined using a correlation between a plurality of first incident information items issued during a third period and a track record of an incident trend based on a plurality of second incident information items issued during a fourth period, wherein each of the third period and the fourth period is earlier than the current point in time, and the fourth period being later than the third period.

27. The information notification method according to claim 1, further comprising: determining a total amount of damage caused by a cyber attack during the second period by using a correlation between a plurality of first incident information items issued during a third period and a track record of a total amount of damage based on a plurality of second incident information items issued during a fourth period, wherein each of the third period and the fourth period is earlier than the current point in time, and the fourth period being later than the third period.

28. An information notification device comprising: a processor; and memory connected to the processor, wherein by using the memory, the processor operates as: an obtainer that obtains a plurality of incident information items on a plurality of incidents caused by cyber attacks on a

monitoring target during a first period, wherein the first period is earlier than a current point in time, and wherein the monitoring target is a vehicle capable of movement; a determiner that determines, based on the plurality of incident information items obtained, for each type of the plurality of incidents: an incident trend of incidents of the type in the monitoring target during a second period; and a priority of a provisional response, which is a response to stop a function involving a vulnerability or to stop a function, or a permanent response, which is a response to execute a software update to overcome the vulnerability to an incident of the type, wherein the second period is later than the current point in time; and a presenter that presents the incident trend and the priority that are determined for each type of the plurality of incidents and are used to perform an operation of the vehicle.
